higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CHEMISTRY
NQF LEVEL 3

4 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1 False
1.2 True
1.3 True
1.4 False
1.5 False

(5 × 1) **[5]**

QUESTION2

2.1 Guano
2.2 Brine
2.3 Additives
2.4 Gypsum
2.5 Cooking liquor

(5 × 1) **[5]**

QUESTION 3

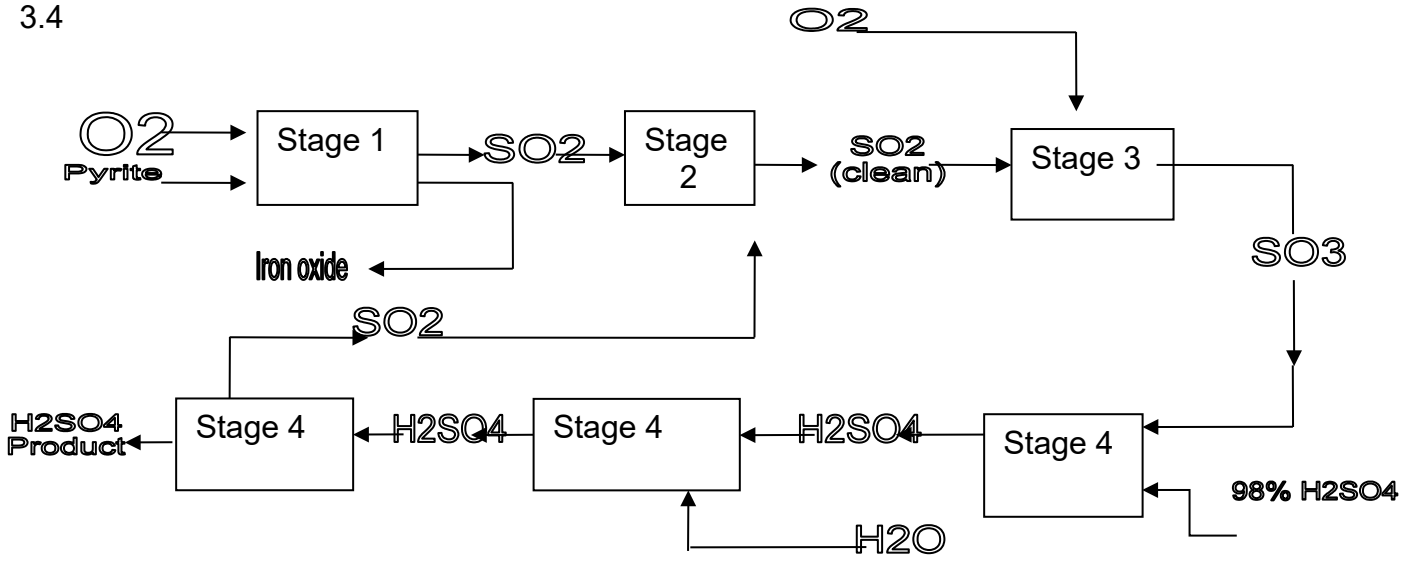
3.1
- They are easy to use
- Nutrients in liquid fertilizers are spread and distributed through the soil easily
- Liquid fertilisers does not need moisture to break it down
- Can be applied whenever the nutrients are needed (4)

3.2
- Sulphur
- Calcium
- Magnesium (3)

3.3
- Used in the agricultural industry as fertilisers
- Added to animal feed to promote healthy growth
- Used as a raw material during the production of plastics and foams
- Also used during the production of adhesives (4)

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3.4



(13)

[24]

QUESTION 4

4.1 4.1.1 Calcium oxide or CaOS
4.1.2 Calcium hydroxide or $Ca(OH)_2$
4.1.3 Calcium carbonate or $CaCO_3$

(3 × 1) (3)

4.2
- Used to preserve food
- Used to improve the flavor of food
- Salt is also used as a cleaning agent
- Manufacture of sodium hydroxide
- Manufacture of chlorine
- Manufacture of sodium carbonate (Any 5 × 1) (5)

4.3
- Sea or ocean water
- Salt water lakes
- Underground salt water pools (3)

4.4 $2NH_4C\ell(s) + Ca(OH)_2(s) \rightarrow CaC\ell\ (s) + 2NH_3(g) + 2H_2O(\ell)$

NOTE: THREE Marks for the products and THREE marks for balancing the equation correctly. (6)

4.5
- It is fire resistance
- It has high strength
- It is durable
- Can be moulded into any shape (Any 3 × 1) (3)

4.6 Strip mining system is used to extract large shallow coal or ore deposits.✓ Trenches are excavated to remove the overburden covering of coal or ore.✓ The layer covering the ore or coal is called the roof. Buckets wheel excavators are used to remove the overburden to a dump.✓ Bucket wheel excavator dumps the extracted ore onto the belt loader connected to the conveyor belt which transported ore to the ground level.✓ (4)

4.7
- Mining of raw materials
- Separation and mixing of raw materials
- Smelting
- Slag granule formation
- Grinding and blending of the slag (5)

4.8
- Rock salt
- Brine (2)

[31]

QUESTION 5

5.1 5.1.1 Soap is made from natural ingredients like vegetables or animal fats and an alkali (sodium hydroxide or caustic soda),✓ whereas detergents are made from synthetic substances like foaming agents, surfactants, fillers and additives.✓ (2)

5.1.2
- Washing powder
- Dishwashing powder
- Dishwashing liquids
- Soap-free liquid cleansers (Any 2 × 1) (2)

5.1.3
- Saponification
- Lye separation (spent lye)
- Soap washing
- Lye separation (half-spent lye)
- Neutralisation
- Drying (Any 6 × 1) (6)

5.2 Fillers are cheap substances that are added to detergents to increase the quantity of the detergents.✓ They also have other secondary functions like pH adjustment or corrosion resistance.✓ (2)

5.3 It is a natural compound which is non-toxic and safe to use as pest control.✓ Boric acid does not kill insects on contact, but it dehydrates them and the insects eventually dry up and die.✓ It also stops the insects' metabolism form functioning properly.✓ (3)

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5.4
- They improve livestock and crop production
- They also improve livestock and crop quality
- Invasion of pests is controlled (Any 2 × 1) (2)

[17]

QUESTION 6

6.1 Bagasse is the fibre that is left after the juices have been squeezed out of sugar cane. (2)

6.2 6.2.1
- Chip bins
- Outdoor chip piles

6.2.2
- Attack by micro-organisms
- Degradation of the cellulose and hemicellulose which result in wood loss

(2 × 2) (4)

6.3
- Entry
- Sheet formation
- Induced static drainage
- Vacuum drainage (4)

6.4 It is the coating and treating of the surface of paper or board to improve the quality, smoothness and the gloss of the paper✓ for the purpose of appearance and printability.✓ It involves the application of an even layer of pigment on to the surface of the sheet.✓ (3)

6.5 A – Round wood
B – Bull screen
C – Primary pressure screens
D – Thickener
E – Mechanical pulp (5)

[18]

TOTAL: 100

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